Library management System.

class

1- Book
2- member
3- Librarian
4- Loan
5- Fine
6- Reservation
7- membership
8- Author.

1- Book . Attributes (Data)	Methods (Behaviour)
• Book ID : Int	• checkAvailability ()
• title : string	• updateBookDetails ()
• author : string	• markAsBorroweed ()
• publisher : string	• markAsReturneed ()
• isAvailable : boolean	

2- member Attributes (Data)	methods (Behaviour)
• memberID : int	• registerMember ()
• name : string	• borrowBook ()
• email : string	• returnBook ()
• phone : sbring	• reserveBook ()
• membershipstatus : string	

→ Loan.

Attributes (Data)	methods (Behaviour)
• LoanID : int	• createLoan ()
• issueDate : Date	• calculateOverdueDays ()
• dueDate : Date	• checkOverdue ()
• returnDate : Date	• closeLoan ()
• LoanStatus : String	

Sentences on Abstraction Decision.

Abstraction means showing only the important details of an object and ignoring unnecessary details.

1- In the Library management system, I included the book title, book ID, author, and availability status because they are important for managing books.

2- I ignored the book's color, weight, paper quality and printing details because they are not necessary for basic borrowing and returning operations.

3- for members I included their ID, name and membership status but ignored unrelated personal details.

Date .....
S M T W TH F S

CRYSTAL

Identify one "Invented" Software object.

Loan Record ID

Generator is a software object that automatically creates a unique ID for every book-borrowing transaction.

example:

When a member borrows a book, the software generates an ID such as LN-1001, then LN-1002 for the next loan.

Why is it use full?

It helps the system identify track and manage each borrowing transaction without confusing one loan with another.